COUNTEREXAMPLES TO THE EQUALITY CLAUSE IN CONJECTURE 20 OF CHEN–GUO–LI–WANG

Abstract. Chen, Guo, Li, and Wang conjectured an upper bound for the product of corresponding Laplacian eigenvalues of a graph and its complement, together with an equality characterization by a specified join construction. We show that the equality characterization is false. For every integer $t \geq 2$, the graph $K_{2t,2t}$ attains equality at $n = 4t$ and $k = 3t$, although the proposed equality family is empty for these parameters. In particular, $K_{4,4}$ gives an eight-vertex counterexample. The product inequality itself is not contradicted.

Let G be a simple graph on n vertices, write $\overline{G}$ for its complement, and order its Laplacian eigenvalues as

$$\mu_1(G) \geq \mu_2(G) \geq \cdots \geq \mu_n(G) = 0.$$

For graphs X and Y, write $X \vee Y$ for their join. Chen, Guo, Li, and Wang [1, Conjecture 20] conjectured that, for every integer k with $1 \leq k \leq 3n/4$,

$$\mu_k(G)\mu_k(\overline{G}) \leq n(n-k),$$

with equality if and only if G or $\overline{G}$ is isomorphic to $K_k \vee H$, where H is a disconnected graph on $n-k$ vertices with at least $k+1$ connected components.

They proved the inequality for $n/2 \leq k \leq 3n/4$ in [1, Theorem 18]. Their argument gives

$$\mu_k(G)\mu_k(\overline{G}) \leq \frac{n^2}{4} \leq n(n-k),$$

so equality in this range can occur only at the endpoint $k = 3n/4$. The following family shows that the proposed characterization omits equality cases at that endpoint.

Theorem 1. *For every integer $t \geq 2$, set*

$$n = 4t, \qquad k = 3t, \qquad G = K_{2t,2t}.$$

Then

$$\mu_k(G)\mu_k(\overline{G}) = n(n-k),$$

but neither G nor $\overline{G}$ belongs to the equality family stated in Conjecture 20.

Proof. Put $m = 2t$. Since $\overline{K_{m,m}} \cong 2K_m$, where $2K_m$ denotes the disjoint union of two copies of K_m, the Laplacian spectra are

$$\mathrm{Spec}_L(K_{m,m}) = \{2m, m^{(2m-2)}, 0\}, \qquad \mathrm{Spec}_L(2K_m) = \{m^{(2m-2)}, 0^{(2)}\},$$

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where superscripts denote multiplicities. Substituting $m = 2t$ gives

$$\mathrm{Spec}_L(G) = \{4t, (2t)^{(4t-2)}, 0\}, \qquad \mathrm{Spec}_L(\overline{G}) = \{(2t)^{(4t-2)}, 0^{(2)}\}.$$

Because $t \geq 2$, we have $3t \leq 4t - 2$, and therefore

$$\mu_{3t}(G) = \mu_{3t}(\overline{G}) = 2t.$$

Consequently,

$$\mu_k(G)\mu_k(\overline{G}) = 4t^2 = (4t)(4t - 3t) = n(n - k).$$

For these parameters, a graph H in the proposed equality family would have $n - k = t$ vertices and at least $k + 1 = 3t + 1$ connected components. This is impossible, since a graph on t vertices has at most t connected components. □

Taking $t = 2$ gives

$$(n, k, G) = (8, 6, K_{4,4}), \qquad \overline{G} \cong 2K_4,$$

and

$$\mu_6(K_{4,4})\mu_6(2K_4) = 4 \cdot 4 = 16 = 8(8 - 6).$$

Thus the "only if" direction of the equality statement in Conjecture 20 fails. These examples attain the proposed upper bound and do not violate the product inequality.

Chen, Guo, Li and Wang state ([1, Remark 1], where the conjecture carries the stale draft label "conjecture 3") that they verified Conjecture 20 for all graphs on at most nine vertices. The counterexample above has eight vertices, so that verification cannot have been correct on the equality clause. We generated all graphs on $2 \leq n \leq 9$ vertices with nauty's `geng` and evaluated $\mu_k(G)\mu_k(\overline{G})$ for $1 \leq k \leq \lfloor 3n/4 \rfloor$. Over this range the inequality $\mu_k(G)\mu_k(\overline{G}) \leq n(n - k)$ is never violated, and the equality characterization fails in exactly five cases, all at $n = 8$, $k = 6$. Up to complementation these are $\{K_{4,4}, 2K_4\}$, the complementary pair with graph6 codes `G?zvf_/GQhTUg`, and the self-complementary graph `GCXnf_`. In particular $K_{4,4}$ is a counterexample of minimum order.

References

[1] Q. Chen, J.-M. Guo, W.-J. Li, and Z. Wang, *A conjecture on the Nordhaus–Gaddum product type inequality for Laplacian eigenvalues of a graph*, Electron. J. Combin. **32** (2025), no. 4, Paper P4.37, 13 pp., doi:10.37236/14060.